ZIP — PlayNAC KERNEL Codebase

> *The working code home of the PlayNAC KERNEL. Where the specification becomes executable.*

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What This Repository Is

This repository hosts the **PlayNAC KERNEL codebase** — the executable counterpart to the canonical PDF specification in `PlayNAC-KERNEL`. Source code, configuration, schemas, and deployable artifacts live here.

> The repository is named `ZIP` for historical reasons (originally a container for program ZIP files). It has been **repositioned as the codebase home**.

Companion Repositories

Repository	Role
`PlayNAC-KERNEL`	PDF specification (what the code implements)
`Support-Documentation`	R&D packaging, auxiliary specs
`Proof-of-Work_MD`	Genesis markdown corpus (prior art)
`Gracechain-Meritcoin`	Economic layer (Proof-of-Resonance)
`Workflows`	CI/CD for this codebase

KERNEL Surface Area

The KERNEL exposes subsystems aligned to the specification:

- **Authentication & Identity** — JWT-based, IDIPITIS-aware (Book 2 mapping).
- **PlayNAC Engine** — task/quest completion, achievements, progression.
- **BERA Measurement** — ARI, ERI, **RHC** (Resonant Harmony Cycle), RCI.
- **UBIMIA distribution** — balance, accrual, settlement.
- **Meritcoin accrual** — **Proof-of-Resonance** (not Proof-of-Work).
- **GAIA tracking** — planetary-tier (T1–T8) state toward JUSTUS end-state.
- **SROC settlement** — Smart-Resonant Offset Contracts.
- **SPT enforcement** — Security, Privacy, Trust per `ERES-SPT-2026-001`.

Technical Stack

Layer	Technology
Runtime	Node.js 18+ (Python 3.8+ companion utilities)
Database	PostgreSQL 12+
Cache / queue	Redis 6+
API	Express + Joi validation
Auth	JWT (access + refresh)

Layer	Technology
Rate limiting	Token bucket, per-endpoint

Quick Start

```bash
# Clone
git clone https://github.com/ERES-Institute-for-New-Age-Cybernetics/ZIP.git
cd ZIP

# Install
npm install

# Configure
cp .env.example .env
# -> set JWT_SECRET, REFRESH_TOKEN_SECRET, DB creds

# Run
npm start
```

Canonical Formulae Used by the Code

```
C = R × P / M                              (Cybernetics = Resource × Purpose /
Method)
M × E + C = R                              (Matter × Energy + Constant = Reason)
REAL = (E·M·R) / (T·S)                     (Energy · Matter · Resonance / Time ·
Space)
RCI  = P_Ω_norm × ARI_sys × VibConst       (Resonant Coherence Index, with
Butzbach)
```

MIEVM-Linked Validation

The code is exercised against MIEVM-validated fixtures derived from the specification. Regression behavior should match the KERNEL PDF; divergence is a bug in the code, not the spec.

Security, Privacy, Trust

All code changes are evaluated against the **SPT Triad** defined in `ERES-SPT-2026-001`. See `SECURITY.md` for disclosure process.

Contributing

See `CONTRIBUTING.md` and `CODE_OF_CONDUCT.md`. Contributions are reviewed against the Three Governing Principles.

Citation

```bibtex
@software{sprute_playnac_kernel_code_2026,
  author       = {Sprute, Joseph A.},
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  url          = {https://github.com/ERES-Institute-for-New-Age-Cybernetics/ZIP}
}
```

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Contact

- **Joseph Allen Sprute** (ERES Maestro) — eresmaestro@gmail.com
- ORCID: 0000-0001-9946-3221

> **Three Governing Principles** — Don't hurt yourself. Don't hurt others. Build for generations to come.

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